

Reducing Hallucinations via Citation-Grounded QA with Locally-Hosted Small Language Models

CS505 Natural Language Processing | Final Project Proposal | Boston University

[We will add names here :)]

Project type: Empirical NLP project focused on retrieval-augmented generation (RAG) for hallucination reduction.

Research question: *Within a fully local, privacy preserving RAG system, how much of a sub-7B SLM's hallucination rate is attributable to retrieval quality versus generation capacity and can systematic retrieval improvements make small local models practically reliable for citation-grounded QA over private corpora?*

1 Problem Statement & Motivation

Hallucination generating claims unsupported by source documents is the primary failure mode of LLMs in high stakes settings. Cloud tools like Google's NotebookLM answer questions over private documents with inline citations, but require transmitting the entire corpus to an external server, which may be incompatible with HIPAA/GDPR requirements or internal data-governance policies. **Cloud upload equals data-exfiltration risk.** Sub-7B SLMs (e.g., Phi-3-mini-3.8B [Microsoft, 2024]) can run on a single consumer GPU, but it is unclear whether their hallucination rate stems primarily from *retrieval failures* or *generation failures*. Disentangling the two is the key to principled, cost-effective improvement.

Task definition. Given a corpus $\mathcal{C} = \{d_1, \dots, d_n\}$ and query q , the system produces answer a plus, for every sentence $s_i \in a$, a citation (p_i, s_i) where passage $p_i \in \mathcal{C}$ textually entails s_i . Any sentence without a valid citation is counted as a hallucination.

2 Proposed System

All components run locally; no external API calls are required.

Parsing. PyMuPDF; preserves section headings for structure-aware chunking.

Chunking (ablated). Three strategies: fixed-size (256 tok, 32 overlap), sliding-window (512 tok, 128 overlap), section-aware (boundary-aligned, capped at 512 tokens to prevent oversized chunks).

Retrieval (ablated). Dense-only (FAISS + BGE-small-en-v1.5); BM25 (rank-bm25); hybrid reciprocal-rank fusion (RRF); hybrid + cross-encoder reranking (ms-marco-MiniLM-L-6-v2), top-5 passages.

Generation. Two prompt variants: standard generation (no citation instruction) vs. *citation-forcing* (the model must bracket every claim with an inline citation and assert only what the retrieved passages support). Served via Ollama; ≤ 16 GB VRAM. If Phi-3-mini fails to follow the bracket format consistently, we will fall back to constrained decoding (Guidance or Outlines) to enforce the schema at the token level.

3 Experimental Design

3.1 Conditions

Each condition changes exactly one component (Table 1). Conditions A to E are the committed core; F and G (both local) are stretch goals testing whether retrieval improvements generalize across model families.

3.2 Dataset & Baselines

Dataset: QASPER [Dasigi et al., 2021]—1,585 NLP papers, 5,049 Q&A pairs with gold evidence annotations supporting both retrieval and answer evaluation (official token F1). Evidence annotations are treated as approximate grounding signals; NLI and token overlap together compensate for annotation incompleteness.

Cond.	Chunk	Retrieval	Rerank	Cite-force	Generator
A (base)	Fixed	Dense	×	×	Phi-3-mini
B	Fixed	Hybrid	×	×	Phi-3-mini
C	Section	Hybrid	×	×	Phi-3-mini
D	Section	Hybrid	✓	×	Phi-3-mini
E (best)	Section	Hybrid	✓	✓	Phi-3-mini
F*	Section	Hybrid	✓	✓	Llama-3-8B
G*	Section	Hybrid	✓	✓	Mistral-7B

Table 1: Ablation conditions. *Stretch goals (time permitting).

We use QASPER as a controlled benchmark for citation-grounded QA; our target deployment is private corpora where gold-annotated evaluation of this kind is not feasible.

Baselines: (1) **Parametric-only LM:** Phi-3-mini with no retrieval, establishing the hallucination floor. (2) **BM25-only** sparse keyword retrieval, the simplest retrieval baseline. (3) **Random Retrieval:** five passages sampled uniformly at random, confirming that RRF gains reflect retrieval *quality* rather than the mere presence of context.

3.3 Metrics & Hypothesis

We treat citation precision as the primary measure of faithfulness; all other metrics serve as supporting signals or retrieval diagnostics.

Primary: Citation precision via NLI entailment (TRUE cross-encoder) and hallucination rate; logs % sentences with valid citations and % malformed citation-format outputs per condition. **Secondary:** Token F1 (QASPER official), BERTScore. **Faithfulness signal:** RAGAS [Es et al., 2023] with Llama-3-8B as local judge (Ollama)—supporting signal only, to avoid LLM-judges-LLM circularity. **Retrieval diagnostics:** Recall@5, MRR@10, evidence hit rate. **Validation:** Human evaluation on 20–30 sampled QA pairs as tie-breaker.

We will additionally plot hallucination rate against Recall@5 across conditions and stratify results by whether gold evidence was retrieved, directly attributing failures to retrieval versus generation.

Hypothesis: Retrieval improvements (A→D) will account for the majority of hallucination reduction, with citation-forcing (D→E) providing a smaller incremental gain; the retrieval–hallucination correlation will show a marked increase in faithfulness once Recall@5 crosses a threshold value.

4 Related Work

Lewis et al. [2020] introduced RAG and remains the canonical reference for retrieval-augmented generation; Asai et al. [2024] is the closest prior system, applying RAG to scientific literature synthesis. Our contribution—a controlled ablation on a locally-hosted SLM that disentangles retrieval failures from generation failures—is absent from both works and from the broader attribution evaluation literature.

5 Feasibility & Timeline

Scope. The committed core (A to E on QASPER with Phi-3-mini) builds the retrieval pipeline incrementally—one component per week—and is realistic on a single consumer GPU. Stretch goals (F: Llama-3-8B, G: Mistral-7B) will be attempted in week 5 and deferred to the final report if time is short.

Stack: PyMuPDF, sentence-transformers, FAISS, rank-bm25, Ollama, RAGAS, HuggingFace Transformers; no paid API for core scope.

References

A. Asai et al. OpenScholar: Synthesizing scientific literature with retrieval-augmented language models. *arXiv preprint arXiv:2411.14199*, 2024.

Wk	Milestone
1–2	Parsing, chunking strategies, BM25 + dense baseline (cond. A)
3	Hybrid RRF + reranking; conditions B to D; QASPER eval pipeline
4	Citation-forcing (E); RAGAS + NLI evaluation; error analysis
5	Midway report; begin stretch goals (F, G) if on schedule
6	Final ablation, writeup (ACL format)

Table 2: Six-week schedule. Core scope: weeks 1–4 (conditions A to E).

- P. Dasigi, K. Lo, I. Beltagy, A. Cohan, N. A. Smith, and M. Gardner. A dataset of information-seeking questions and answers anchored in research papers. In *Proceedings of NAACL*, 2021.
- S. Es, J. James, L. Espinosa-Anke, and S. Schockaert. RAGAS: Automated evaluation of retrieval augmented generation. *arXiv preprint arXiv:2309.15217*, 2023.
- P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. arXiv:2005.11401.
- Microsoft. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024.